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Estimating Inhomogeneous Spatiotemporal Background Intensity Functions Using Graphical Dirichlet Processes

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Abstract
Enhancements in seismic monitoring instrumentation have been shown to have an impact on the number of observed earthquakes, since denser networks usually allow the recording of more events. However, phenomena such as strong earthquakes or aseismic transients, including slow slip events, may alter the seismicity rate. In the field of seismology, it is standard practice to model background seismicity as a Poisson process. Based on this idea, this work proposes a model that can incorporate the evolving spatial intensity of Poisson processes over time, which implies that temporal changes in background seismicity are included in the modeling. In recent years, novel methodologies have been developed for quantifying the uncertainty in the estimation of background seismicity in homogeneous cases using Bayesian nonparametric techniques. The present study develops a novel methodology based on graphical Dirichlet processes for incorporating spatial and temporal inhomogeneities in background seismicity. The proposed model is applied to study seismicity in southern Mexico, using recorded data from 2000 to 2015.

Keywords ETAS · Dirichlet process · Bayesian nonparametric · Background seismicity

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1 Introduction

Since its introduction, the epidemic type aftershock sequence (ETAS) model (Ogata 1988) has emerged as one of the most important tools for assessing stochastic seismicity, in which earthquake activity is modeled using a Hawkes process (Hawkes 1971). In this context, it is assumed that mainshocks (or background earthquakes) originate from a Poisson process. Recently, due to the importance of quantifying the uncertainty of estimators, authors such as Ross and Kolev (2022) and Molkenthin et al. (2022) have worked with Bayesian nonparametric approaches to model a temporally homogeneous background intensity function, μ in Eq. (1), using Dirichlet process mixtures (DPMs) (Antoniak 1974) and Gaussian process priors, respectively.

Since the homogeneity of the ETAS model can be affected by tectonic or pore pressure changes (Hainzl and Ogata 2005; Hainzl et al. 2013; Marsan et al. 2013), such phenomena may affect tremor occurrence. Furthermore, these changes may be induced by strong earthquakes (Cruz-Atienza et al. 2021) and may also lead to an increase in seismicity rate (Nishikawa and Ide 2017). Similarly, the presence of slow slip events, which are slow fault slip phenomena that mainly occur along the plate interface, can alter seismic activity (Marsan et al. 2013; Fukuda 2018; Nishikawa and Nishimura 2023; Bañales et al. 2025) and can also lead to periods of microearthquake quiescence (Matthews and Reasenberg 1988).

Moreover, not only can variations in seismicity affect ETAS model estimates, but factors such as short-term aftershock incompleteness (Hainzl 2016; Asayesh et al. 2025) and changes in the cutoff magnitude (Hainzl 2022) can also influence these estimates. These temporary changes in activity emphasize the need for time-inhomogeneous models for the background intensity function, such as the one introduced in this paper.

The contribution of this work is framed in three main aspects. First, a nonparametric Bayesian approach is proposed to estimate the background intensity function based on graphical Dirichlet processes (GDPs) (Chakrabarti et al. 2024), allowing for spatiotemporal inhomogeneities. Second, the background intensity is inferred with temporal variation, that is, $\mu(x, y, t)$, under the GDP framework. In the context of tectonic activity, this formulation is relevant because it may provide insights into the underlying physical processes governing earthquake occurrence. Finally, the analysis is restricted to the background intensity function, since the database compiled by Sawires et al. (2019) is employed, in which aftershocks, foreshocks, and swarms have been removed using the methodology of Gardner and Knopoff (1974), leaving only mainshocks and thus yielding a purely Poissonian intensity.

In the seismological literature, there are two main approaches for distinguishing background events from aftershocks: deterministic spatiotemporal filtering methods, as proposed by Gardner and Knopoff (1974), and stochastic approaches based on the ETAS model of Ogata and Katsura (1988), as in Zhuang et al. (2002) and Schoenberg (2013). Earthquake occurrence is typically modeled as a self-exciting point process. Nevertheless, because the catalog of Sawires et al. (2019) is employed, the present study focuses exclusively on background seismicity rather than on a fully self-exciting process. Irrespective of the catalog used, any assumption about the background intensity function may introduce bias in the estimation of triggering effects.

While the catalog of Sawires et al. (2019) allows the focus to be placed solely on the Poissonian part of the problem, Sect. 2 describes how extensions of the model proposed in Ross and Kolev (2022), based on the ETAS model, can be generated by incorporating hidden variables in a manner similar to that developed in Schoenberg (2013) and Bañales et al. (2025).

In the frequentist context, Li et al. (2020) used piecewise constant functions and kernel density estimators to model an inhomogeneous space-time background activity function within the Hawkes process framework. They assumed that

$$\mu(x, y, t) = \gamma f(x, y) v(t),$$

where $\gamma \in \mathbb{R}^+$, f is a probability density function, and v is a real-valued function. Nevertheless, the assumption that spatial changes can be separated from temporal changes is generally not realistic for earthquake activity.

In the nonparametric Bayesian framework, the estimation of $\mu(x, y, t)$ can be done using the generalized spatial Dirichlet process introduced by Duan et al. (2007). Their approach allows modeling different probability density functions across space and discretized time, where the weights and point masses in the stick-breaking representation (Sethuraman 1994) are assumed to vary smoothly over space and time. However, their formulation requires modeling the weights through an auxiliary threshold scheme based on a collection of independent stationary Gaussian random fields. Since the mass points are kept fixed in the present work, the representation proposed by Chakrabarti et al. (2024) is adopted, which allows the weights to vary through a hierarchical model with Beta distributions.

It is also worth mentioning the work by Fonseca and Ferreira (2017), which deals with the spatiotemporal problem that occurs when point data are not directly observed and only aggregated counts are available. In this context, Kottas and Sansó (2007) proposed a nonparametric Bayesian approach to estimate the time-homogeneous spatial Poisson-process intensity function using Dirichlet processes, which was extended to the inhomogeneous case by Xiao et al. (2015) using dependent Dirichlet processes (DDPs) (Maceachern 1999). If the reader is interested in DDPs and their stick-breaking representation, it is suggested that they consult Quintana et al. (2022).

The main difference between the methodology of Xiao et al. (2015) and the one proposed here lies in the DDP structure: while Xiao et al. (2015) keeps the mixture weights fixed over time and allows the component locations to vary, the locations are instead kept fixed and the weights are allowed to evolve over time, as discussed in Sect. 2. The idea of keeping mixture locations constant over time is reasonable, given that regions where tectonic plates interact require very long periods to undergo significant changes. Therefore, the main seismogenic regions remain stable in location. When a change in seismicity occurs, for example, due to a strong earthquake or a slow slip event (SSE), the weight associated with the mixture component in these regions deviates from zero only during periods near the occurrence of the event. Combined with the fact that GDP requires fewer parameters to be estimated, this approach is particularly attractive in the context of seismology.

The modeling of spatiotemporal point processes is not limited to seismology. In particular, recent works such as Gilardi et al. (2024) and Medialdea et al. (2025) have proposed models based on spatial Cox processes (Diggle et al. 2013) to model ambulance interventions and wildfires. Using a frequentist approach, these models also allow the incorporation of exogenous variables and planar geometries. Also, in the context of self-exciting point processes, Briz-Redón and Mateu (2024) proposed a purely temporal model in which the self-exciting mechanism can be activated and deactivated over time, thereby reducing the underestimation of the productivity parameter in a seismological context.

Finally, the works of Zhuang and Mateu (2019) and Escudero et al. (2025) are mentioned, which focus on modeling crime data using log-Gaussian Cox and Hawkes processes with inhomogeneous separable spatiotemporal background intensities. Their approach contrasts with the non-separable intensity proposed in this work based on a GDP, which highlights the novelty of the approach presented in Sect. 2.

This paper is structured as follows: In Sect. 2, the first-order autoregressive GDP is presented, along with its finite mixture approximation. In Sect. 3, a synthetic example and an application to southern Mexico data, based on the catalog of Sawires et al. (2019), are described. Finally, in Sect. 4, the results are discussed and directions for future work are outlined.

2 Methodology

As noted by Fox et al. (2016), the usual approach for modeling mainshocks (i.e., earthquakes that are not triggered by other earthquakes, unlike aftershocks) is to use an inhomogeneous Poisson process in space, assumed constant over time, with intensity $\mu(x, y)$. The rate of aftershocks is modeled using a triggering function ν depending on the magnitudes of earthquakes, M_i for $i = 1, 2, \ldots, N$, which decay according to the spatial and temporal distance between earthquakes. The intensity function in the ETAS model is given by

$$\lambda^*_{\mu,\nu}(t, x, y|H_t) = \mu(x, y) + \sum_{\{i:t_i<t\}} \nu(t - t_i, x - x_i, y - y_i, M_i), \tag{1}$$

where (x_i, y_i) are the coordinates of the epicenter, and t_i is the time of occurrence of the event i, with $i = 1, 2, \ldots, N$, and H_t denoting the history of earthquake locations, times, and magnitudes up to time t. Physically, $\mu(x, y)$ denotes the background seismicity, which refers to the earthquake rate that is unaffected by the occurrence of any other seismic event. The triggering function ν is described as the decay of an earthquake's ability to generate aftershocks over time (i.e., based on Omori's law (Utsu et al. 1995)). Only earthquakes with magnitude exceeding the completeness threshold M_u are considered; these are assumed to follow the Gutenberg-Richter law (Gutenberg and Richter 1949). The log-likelihood of the model is given by

$$\ell(\mu, \nu) := \sum_{k=1}^{n} \log(\lambda^*_{\mu,\nu}(t_k, x_k, y_k|H_{t_k})) - \int_0^T \iint_D \lambda^*_{\mu,\nu}(t, x, y|H_t)dxdydt, \tag{2}$$

where $(0, T)$ is the observation time interval and $D \subset \mathbb{R}^2$ denotes the observed region.

As discussed in Sect. 1, the catalog from Sawires et al. (2019) allows the focus to be placed solely on the Poissonian intensity function. Therefore, throughout the remainder of this work, an inhomogeneous Poisson process (NHPP) with intensity function defined as

$$\lambda(t, x, y) = \mu(x, y, t), \tag{3}$$

is considered for modeling the background intensity function.

It is important to note that if a sample $X = \{(x_i, y_i, t_i)\}$ is observed, where (x_i, y_i) are observed in a region $D \subset \mathbb{R}^2$, with $i = 1, 2, \ldots, N$ over a time interval $(0, T)$ from an NHPP, the likelihood (Daley and Vere-Jones 2003) is given by

$$L(X|\mu) = e^{-\int_0^T \iint_D \mu(x,y,t)dtdxdy} \prod_{i=1}^{N} \mu(x_i, y_i, t_i), \tag{4}$$

which, in general, is computationally expensive, as it requires solving a triple integral. Consequently, authors such as Kottas and Sansó (2007), Zhuang et al. (2002), and Ross and Kolev (2022) have assumed that in the stationary case, $\mu(x, y)$ is a bounded function. This form allows it to be modeled as

$$\mu(x, y) = \gamma f(x, y),$$

where γ is in $\mathbb{R}^+$, and $f(x, y)$ is a density function. Additionally, if the domain of $f(x, y)$ is D, then the likelihood for the homogeneous case (in time), Eq. (4), can be rewritten as

$$L(X|\mu) = e^{-\gamma T} \gamma^N \prod_{i=1}^{N} f(x_i, y_i).$$

This significantly reduces its computational cost, and the idea of using the Dirichlet process to model μ, as in Kottas and Sansó (2007) or Ross and Kolev (2022), emerges due to the independent and identically distributed (i.i.d.) sampling of (x_i, y_i), if γ were known.

As discussed in Sect. 1, following the idea of writing $\mu(x, y) = \gamma f(x, y)$, authors such as Li and Cui (2024) and Xiao et al. (2015) have proposed defining μ as a spatial piecewise constant function over time. That is, if a partition of the interval $(0, T)$ is given by the times $0 = S_1, S_2, \ldots, S_{p+1} = T$, then μ can be defined as

$$\mu(x, y, t) = \sum_{p=1}^{P} \gamma_p f_p(x, y) \mathbb{1}(S_p < t < S_{p+1}), \tag{5}$$

where $\mathbb{1}(\cdot)$ is the indicator function, which is the definition of μ used in this work. This definition aims to account for inhomogeneities in both space and time, but also

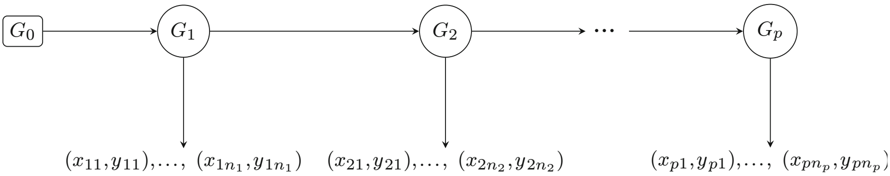

Fig. 1 Directed acyclic graph representation of the first-order graphical Dirichlet process illustrating temporal dependence between mixture distributions

allows the likelihood expressed in Eq. (4) to be rewritten as

$$L(X|\mu) = \prod_{p=1}^{P} \left(e^{-\gamma_p |I_p|} \gamma^{n_p} \prod_{i=1}^{n_p} f_p(x_{p,i}, y_{p,i}) \right), \tag{6}$$

where $I_p = (S_p, S_{p+1})$, $|I_p|$ is the width of the partition, and $X_p = \{(x_{p,i}, y_{p,i})\}$ with $i = 1, \ldots, n_p$ is the subsample of X of elements where t_i is in I_p for $p = 1, \ldots, P$.

Following Eq. (5), non-exchangeable subsamples of X are obtained for each partition, and this provides a reasonable model when dependence is assumed. The temporal dependency used in this work can be described as a particular case of GDP, for which the directed acyclic graph is shown in Fig. 1. The GDP (Chakrabarti et al. 2024) allows for working with a more general directed acyclic graph (DAG). Nevertheless, for the present purposes, this formulation is sufficient. It also coincides with structures previously studied in the nested Chinese restaurant process (Blei et al. 2010) and the nested hierarchical Dirichlet processes (Paisley et al. 2014), both of which are encompassed within the GDP framework.

As mentioned by Müller et al. (2015), Dirichlet processes are discrete with probability 1; hence, DPMs are commonly used to obtain smooth estimators. This approach is generalized in Chakrabarti et al. (2024) to the GDP, where it is proved that the structure presented in Fig. 1, used to define a GDP mixture, is

$$G_1|\alpha_1, G_0 \sim DP(\alpha_1, G_0),$$
$$G_p|\alpha_p, G_{p-1} \sim DP(\alpha_p, G_{p-1}),$$

and for each group p, it holds that

$$\theta_{p,i}|G_p \overset{\text{ind}}{\sim} G_p,$$
$$(x_{p,i}, y_{p,i})|\theta_{p,i} \sim F(\theta_{p,i}),$$

which, by the stick-breaking representation, can be recovered as the limit of the hierarchical finite mixture model used in Eq. (7) when the number of mixture elements L tends to infinity.

Considering the γ_p variables, a priori independent of all G_p processes and Eq. (5), the prior distribution of $\mu(x, y, t)$ in this work, is given by

$$
\begin{aligned}
\alpha_1|\alpha_0 &\sim \text{Gamma}(\alpha_0, 1),\\
\beta_1|\alpha_1 &\sim \text{Dir}\Big(\frac{\alpha_1}{L}, \dots, \frac{\alpha_1}{L}\Big),\\
\alpha_p|\alpha_{p-1} &\sim \text{Gamma}(\alpha_{p-1}, 1),\\
\beta_p|\alpha_p, \beta_{p-1} &\sim \text{Dir}(\alpha_p(\beta_{p-1,1}, \beta_{p-1,2}, \dots, \beta_{p-1,L})),\\
\psi_l|G_0 &\sim G_0,\\
z_{p,i}|\beta_p &\sim \text{Cat}(1:L, \beta_p),\\
\gamma_p &\sim \text{Gamma}(\gamma_0 k, k),\\
\{(x_{p,i}, y_{p,i})|z_{p,i}, \{\psi_l\}_{l=1}^L, \gamma_p\}_{i=1}^{n_p} &\sim \mathcal{F}(\{z_{p,i}\}_{i=1}^{n_p}, \{\psi_l\}_{l=1}^L, \gamma_p) \text{ for all } p \text{ in } \{1, 2, \dots, P\},
\end{aligned} \tag{7}
$$

where $\{(x_{p,i}, y_{p,i})|z_{p,i}, \{\psi_l\}_{l=1}^L, \gamma_p\}_{i=1}^{n_p} \sim \mathcal{F}(\{z_{p,i}\}_{i=1}^{n_p}, \{\psi_l\}_{l=1}^L, \gamma_p)$ has the probability density function

$$
f((x_{p,1}, y_{p,1}), \dots, (x_{p,n_p}, y_{p,n_p})) = e^{-\gamma_p|I_p|}\gamma_p^{n_p} \prod_{i=1}^{n_p} \Big(\prod_{l=1}^{L} \phi_{\psi_l}(x_{p,i}, y_{p,i})^{\mathbb{1}(z_{p,i}=l)} \Big),
$$

and, as previously introduced in Eq. (6), the likelihood is given by

$$
f(X|\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}, \boldsymbol{\psi}, \mathbf{Z}) = \prod_{p=1}^{P} \Big\{ e^{-\gamma_p|I_p|}\gamma_p^{n_p} \prod_{i=1}^{n_p} \Big(\prod_{l=1}^{L} \phi_{\psi_l}(x_{p,i}, y_{p,i})^{\mathbb{1}(z_{p,i}=l)} \Big) \Big\}.
$$

The functions $\phi_{\mu_l, \Sigma_l^{-1}}$, for $l = 1, 2, \dots, L$, are assumed to be Gaussian probability densities with mean μ_l and covariance matrix Σ_l. For notational convenience, let $\psi_l = (\mu_l, \Sigma_l^{-1})$ and denote these densities by ϕ_{ψ_l}. Although this assumption may violate the requirement that f has support D, following Ross and Kolev (2022), the probability mass outside D is considered negligible. Therefore, Eq. (6) remains valid.

The main reason for assuming that f_{ψ_l} is Gaussian is to obtain the full conditional distribution of ψ_l via conjugate analysis, which allows its straightforward incorporation into a Markov chain Monte Carlo (MCMC) scheme. It is worth mentioning that Kottas and Sansó (2007) proposed a bivariate beta distribution to ensure bounded support over D. Nevertheless, following the aforementioned reasoning, the approach proposed by Ross and Kolev (2022) is followed.

For all the examples, $G_0 \sim \text{NIW}(\mu_0, \eta, \Sigma_0, \nu)$, where NIW denotes a normal inverse Wishart distribution; this density is also denoted by $f_{\text{NIW}}(\cdot)$. While the prior distribution reflects uncertainty about the parameters before observing the data, this uncertainty is updated once the sample is observed via Bayes' theorem, as detailed in Bernardo et al. (1994), which defines the posterior distribution, that is,

$$
f(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}, \boldsymbol{\psi}, \mathbf{Z}|X) = \frac{f(X|\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}, \boldsymbol{\psi}, \mathbf{Z}) f(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}, \boldsymbol{\psi}, \mathbf{Z})}{f(X)}.
$$

Since the denominator is only a normalizing constant and does not depend on the parameters of interest, the posterior distribution is fully characterized by the expression

$$
\begin{aligned}
f(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}, \boldsymbol{\psi}, \mathbf{Z}|X) &\propto f(X|\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}, \boldsymbol{\psi}, \mathbf{Z}) f(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}, \boldsymbol{\psi}, \mathbf{Z}) \qquad (8)\\
&= \alpha_1^{\alpha_0-1} e^{-\alpha_1} \frac{\prod_{l=1}^{L} \beta_{1l}^{\frac{\alpha_1}{L}-1}}{\mathrm{B}(\frac{\alpha_1}{L}, \frac{\alpha_1}{L}, \ldots, \frac{\alpha_1}{L})} \Big(\prod_{p=1}^{P-1} \frac{\alpha_{p+1}^{\alpha_p-1} e^{-\alpha_{p+1}}}{\Gamma(\alpha_p)} \Big)\\
&\times \Big(\prod_{p=2}^{P} \frac{\prod_{l=1}^{L} \beta_{pl}^{\alpha_p \beta_{p-1,l}-1}}{\mathrm{B}(\alpha_p \beta_{p-1,1}, \ldots, \alpha_p \beta_{p-1,L})} \Big) \Big(\prod_{l=1}^{L} f_{\mathrm{NIW}}(\psi_l) \Big)\\
&\times \Big(\prod_{p=1}^{P} \prod_{i=1}^{n_p} \prod_{l=1}^{L} \beta_{pl}^{\mathbb{1}(z_{pi}=l)} \Big) \Big(\prod_{p=1}^{P} \gamma_p^{\gamma_0 k-1} e^{-\gamma_p k} \Big)\\
&\times \prod_{p=1}^{P} e^{-\gamma_p |I_p|} \gamma_p^{n_p} \prod_{i=1}^{n_p} \Big(\prod_{l=1}^{L} \phi_{\psi_l}(x_{p,i}, y_{p,i})^{\mathbb{1}(z_{p,i}=l)} \Big),
\end{aligned}
$$

where

$$
\begin{aligned}
\boldsymbol{\alpha} &= (\alpha_1, \ldots, \alpha_P),\\
\boldsymbol{\gamma} &= (\gamma_1, \ldots, \gamma_P),\\
\boldsymbol{\psi} &= (\psi_1, \ldots, \psi_L),\\
\mathbf{Z} &= (z_{1,1}, \ldots, z_{1,n_1}, z_{2,1}, \ldots, z_{2,n_2}, \ldots, z_{P,n_P}),\\
\boldsymbol{\beta} &= \begin{pmatrix} , \beta_{11} & \ldots & \beta_{1L} \\ \vdots & \ddots & \vdots \\ \beta_{P1} & \ldots & \beta_{PL} = \end{pmatrix} = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_P \end{pmatrix}.
\end{aligned}
$$

In order to facilitate efficient sampling from the posterior distribution, the full conditional distributions of the model parameters are derived using the definition of conditional probability. In particular, the full conditional distribution of γ_1 is given by

$$
f(\gamma_1|\alpha, \boldsymbol{\beta}, \boldsymbol{\gamma}_{-1}, \boldsymbol{\psi}, \mathbf{Z}, X) = \frac{f(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}, \boldsymbol{\psi}, \mathbf{Z}|X)}{f(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}_{-1}, \boldsymbol{\psi}, \mathbf{Z}, X)} \propto f(\boldsymbol{\alpha}, \boldsymbol{\beta}, \boldsymbol{\gamma}, \boldsymbol{\psi}, \mathbf{Z}|X),
$$

where $\boldsymbol{\gamma}_{-1}$ denotes the vector $\boldsymbol{\gamma}$ without its first component. This means that the full conditional distribution of a parameter is proportional to the posterior distribution, treating the remaining parameters as constants. Therefore, it is easy to see that the full conditional for γ_p, for all $p = 1, \ldots, P$, is distributed as Gamma$(\gamma_0 k + n_p, k + |I_p|)$.

For ψ_l with $l = 1, \ldots, L$ the full conditional is given by

$$\pi(\psi_l|\cdot) \propto \prod_{p=1}^{P}\prod_{i=1}^{n_p}\left(\prod_{l=1}^{L}\phi_{\psi_l}(x_{p,i}, y_{p,i})^{\mathbb{1}(z_{p,i}=l)}\right) f_{\mathrm{NIW}}(\psi_l),$$

which, by conjugacy, results in an NIW posterior for ψ_l, and implies that $(x_{p,i}, y_{p,i})$ are i.i.d. Gaussian conditional on ψ_l. This is a well-known result from conjugate analysis, which can be found in Murphy (2022). The resulting distribution is $\mathrm{NIW}(\mu_l, \eta_l, \Sigma_l, \nu_l)$, where

$$\begin{aligned}
\mu_l &= \frac{\eta\mu_0 + m_l\widehat{\mu}_{xy}}{\eta + m_l},\\
\eta_l &= \eta + m_l,\\
\Sigma_l &= \Sigma_0 + S + \frac{\eta m_l}{\eta + m_l}(\widehat{\mu}_{xy} - \mu_0)(\widehat{\mu}_{xy} - \mu_0)^T,\\
\nu_l &= \nu + m_l,
\end{aligned}$$

defined by

$$\begin{aligned}
m_l &= \sum_{p=1}^{P}\sum_{i=1}^{n_p}\mathbb{1}(z_{p,i} = l),\\
\widehat{\mu}_{xy} &= \frac{\sum_{p=1}^{P}\sum_{i=1}^{n_p}(x_{p,i}, y_{p,i})\mathbb{1}(z_{p,i} = l)}{m_l},\\
S &= \sum_{p=1}^{P}\sum_{i=1}^{n_p}((x_{p,i}, y_{p,i}) - \widehat{\mu}_{xy})((x_{p,i}, y_{p,i}) - \widehat{\mu}_{xy})^T\mathbb{1}(z_{p,i} = l).
\end{aligned}$$

It is important to note that ψ_l, for $l = 1, \ldots, L$, are not independent in the posterior, as can be seen from Eq. (8), even though they were assumed independent a priori. Furthermore, their full conditional means incorporate the samples $(x_{p,i}, y_{p,i})$ for all $i = 1, \ldots, n_p$ and $p = 1, \ldots, P$, thus using information from the observed earthquakes across all partitions.

In the case of $\boldsymbol{\alpha}$, the following result is obtained

$$\begin{aligned}
\pi(\alpha_1|\cdot) &\propto \frac{e^{-\alpha_1}\alpha_1^{\alpha_0-1}\alpha_2^{\alpha_1}}{\Gamma(\alpha_1)}\frac{\prod_{l=1}^{L}\beta_{1l}^{\frac{\alpha_1}{L}}}{\mathrm{B}(\frac{\alpha_1}{L}, \frac{\alpha_1}{L}, \ldots, \frac{\alpha_1}{L})},\\
\pi(\alpha_P|\cdot) &\propto e^{-\alpha_P}\alpha_P^{\alpha_{P-1}-1}\frac{\prod_{l=1}^{L}\beta_{Pl}^{\alpha_P\beta_{P-1,l}}}{\mathrm{B}(\alpha_P\beta_{P-1,1}, \ldots, \alpha_P\beta_{P-1,L})},
\end{aligned}$$

and for all $p = 2, \dots, P-1$,

$$\pi(\alpha_p|\cdot) \propto \frac{e^{-\alpha_p}\alpha_p^{\alpha_{p-1}-1}\alpha_{p+1}^{\alpha_p}}{\Gamma(\alpha_p)} \frac{\prod_{l=1}^{L}\beta_{pl}^{\alpha_p\beta_{p-1,l}}}{\mathrm{B}(\alpha_p\beta_{p-1,1},\dots,\alpha_p\beta_{p-1,L})}.$$

Also, for $\boldsymbol{\beta}$, the following is obtained

$$\pi(\beta_1|\cdot) \propto \prod_{l=1}^{L}\beta_{1l}^{m_{1l}+\frac{\alpha_1}{L}-1} \frac{\prod_{l=1}^{L}\beta_{2l}^{\alpha_2\beta_{1,l}}}{\mathrm{B}(\alpha_2\beta_{1,1},\dots,\alpha_2\beta_{1,L})},$$

in a similar way, for $p = 1, 2, \dots, P-1$

$$\pi(\beta_p|\cdot) \propto \prod_{l=1}^{L}\beta_{pl}^{m_{pl}+\alpha_p\beta_{p-1,l}-1} \frac{\prod_{l=1}^{L}\beta_{p+1,l}^{\alpha_{p+1}\beta_{p,l}}}{\mathrm{B}(\alpha_{p+1}\beta_{p,1},\dots,\alpha_{p+1}\beta_{p,L})},$$

where $m_{pl} = \sum_{i=1}^{n_p}\mathbb{1}(z_{p,i} = l)$. Meanwhile, for $p = P$, since the term $p+1$ does not exist, the following distribution is obtained

$$\beta_P|\cdot \sim \text{Dirichlet}(m_{P1} + \alpha_P\beta_{P-1,1}, \dots, m_{PL} + \alpha_P\beta_{P-1,L}).$$

The full conditional distribution of β_p, for $p = 2, \dots, P-1$, depends on the neighboring vectors β_{p-1} and β_{p+1}. Thus, although an autoregressive structure is specified a priori, the resulting posterior dependence reinforces temporal coupling between adjacent coefficients. In combination with $\boldsymbol{\alpha}$, this structure allows the model to capture smooth temporal behavior when $\boldsymbol{\alpha}$ is large, while permitting more abrupt changes in the density when $\boldsymbol{\alpha}$ is close to zero.

A limitation of this approach is that, as the norm of the time partition tends to zero, the number of observations within each interval becomes very small, which complicates estimation. However, this issue can be mitigated by imposing an informative prior on $\boldsymbol{\alpha}$, thereby enforcing a smoothness structure on the spatiotemporal evolution. This regularization is particularly useful in seismic catalogs where large-magnitude slow slip events (SSEs) are observed without the occurrence of strong earthquakes. Finally,

$$\mathbb{P}(z_{p,i} = l|\cdot) \propto \beta_{pl}\phi_{\psi_l}(x_{p,i}, y_{p,i}),$$

for $i = 1, 2, \dots, n_p$.

The autoregressive GDP of order 1 was selected for simplicity. However, as detailed in Chakrabarti et al. (2024), this framework can be extended to higher orders. Furthermore, as can be seen from the posterior distribution, the parameters are not independent, even for noncontiguous time instants.

As discussed in Chakrabarti et al. (2024), the full conditional distributions of α_i, $i = 1, \ldots, P$, and β_p, for $p = 1, \ldots, P-1$, are not standard distributions, and constitute the main bottleneck in the MCMC. Since all α_i are positive random variables, a random walk step is proposed to explore them. According to Chakrabarti et al. (2024), proposals based on the SALTSampler, introduced by Director et al. (2017), are used to explore β_p. Then, a hybrid MCMC (Robert et al. 1999) is employed to sample from the posterior distribution; the code is available at github.com/isaiasmanuel/NHPP.

Since Veen and Schoenberg (2008), the use of hidden variables in the ETAS model has increased in popularity due to the computational cost and numerical instability of direct likelihood optimization, originally proposed by Zhuang et al. (2002). The hidden variables included are

$$\chi_{ii} = \begin{cases} 1, & \text{if earthquake } i \text{ is a background event} \\ 0, & \text{otherwise} \end{cases}, \tag{9}$$

$$\chi_{ij} = \begin{cases} 1, & \text{if earthquake } i \text{ is an aftershock of } j \\ 0, & \text{otherwise} \end{cases}. \tag{10}$$

These variables represent the branching structure of earthquakes, and the log-likelihood of the ETAS model that incorporates them is given by

$$\ell_c\Big(\mu, \eta, \{\chi_{ii}\}_{i=1}^N, \{\chi_{ij}\}_{(i,j)\in I^2}\Big) = \ell_{\mathrm{O}}^*(\mu, \{\chi_{ii}\}_{i=1}^N) + \ell_{\mathrm{I}}^*(\nu, \{\chi_{ij}\}_{(i,j)\in I^2}),$$

where I^2 denotes $\{1, 2, \ldots, N\} \times \{1, 2, \ldots, N\}$, with $\times$ denoting the Cartesian product, $\ell_{\mathrm{O}}(\cdot)$ is the log-likelihood due to offspring, and $\ell_{\mathrm{I}}(\cdot)$ is the log-likelihood due to immigrants. The two components are given by

$$\ell_{\mathrm{I}}^*(\mu, \{\chi_{ii}\}_{i=1}^N) = \sum_{i=1}^n \chi_{ii} \log(\mu(x_i, y_i)) - \int_0^T \iint_S \mu(x, y, t)dxdydt, \tag{11}$$

$$\begin{aligned}\ell_{\mathrm{O}}^*(\nu, \{\chi_{ij}\}_{(i,j)\in I^2}) = \sum_{j=1}^n \Big[\sum_{i>j} \chi_{ij} \log\Big(\nu(t_i - t_j, x_i - x_j, y_i - y_j, M_j)\Big) \\ - \int_{t_j}^T \iint_S \nu(t - t_j, x - x_j, y - y_j, M_j)dxdydt\Big].\end{aligned} \tag{12}$$

Within this framework, authors such as Veen and Schoenberg (2008) and Li et al. (2020) have used the EM algorithm for the ETAS model. Meanwhile, Molkenthin et al. (2022) and Ross and Kolev (2022), among others, have introduced Bayesian hierarchical schemes based on Gibbs proposals. Only the Poissonian term of the model presented in this section can be incorporated into the ETAS framework by redefining ℓ_{I}^* as

$$\ell_{\mathrm{I}}^{*}(\mu, \{\chi_{ii}\}_{i=1}^{N}) = \sum_{p=1}^{P} \Big(\sum_{i=1}^{n} \chi_{ii} \log(\mu^{p}(x_i, y_i, t_i)) - \int_0^T \iint_S \mu^{p}(x, y, t) dxdydt \Big).$$

However, in the examples of Sect. 3, this evaluation is not necessary to estimate the background intensity function, because the features of the Sawires et al. (2019) catalog allow the assumption that $\chi_{ij} = 0$ for all $i \neq j$ and $\chi_{ii} = 1$ for all i.

3 Numerical Experiments

3.1 Simulated Data

To verify the performance of the model presented in Sect. 2, it is applied to a synthetic catalog of mainshocks with $D = (-5, 10) \times (-5, 10)$ and $t \in (0, 10)$, using the intensity function

$$\lambda(x, y, t) = \Big(50\mathbb{1}(t < 5) + 100\mathbb{1}(t \geq 5)\Big)\Big(h(t)g_1(x, y) + (1 - h(t))g_2(x, y)\Big), \quad (13)$$

where $\mathbb{1}(\cdot)$ denotes the indicator function, and

$$\begin{aligned} h(t) &= \frac{1}{1 + e^{-\frac{t-T}{2}}}, \\ g_1(x, y) &= \frac{2}{3}\phi_{(0,0),\mathbf{1}_{2\times 2}}(x, y) + \frac{1}{3}\phi_{(2,2),\mathbf{1}_{2\times 2}}(x, y), \\ g_2(x, y) &= \frac{2}{3}\phi_{(6,2),\mathbf{1}_{2\times 2}}(x, y) + \frac{1}{3}\phi_{(4,6),\mathbf{1}_{2\times 2}}(x, y). \end{aligned}$$

The idea behind the proposed λ function is to construct a spatiotemporal intensity function capable of capturing the typical challenges observed in earthquake catalogs. For example, Bañales et al. (2025) show that SSEs can modify seismicity. In southern Mexico, SSEs can last longer than 1 year, which motivates the function h in Eq. (13) and allows for a smooth transition between modes. Additionally, the number of recorded earthquakes may increase due to changes in instrumentation or aseismic transients, as discussed in Sect. 1. In Mexico, improvements in instrumentation, as described in Vergnolle et al. (2010) and Cruz-Atienza et al. (2018), have increased the number of recorded earthquakes. This non-smooth change is reflected in Eq. (13) by changing the productivity parameter from 50 to 100 at time 5.

In this example, the Normal inverse Wishart distribution is considered with $\mu_0 = (1, 1)$, where the vector $(1, 1)$ was chosen arbitrarily inside D, $\Sigma_0 = \mathbb{I}_2$, $\nu = 3$, and $\eta = 0.1$. Here, $\mathbb{I}_2$ denotes the 2×2 identity matrix. The idea is to use a non-informative prior distribution for the parameters in the mixture. For all the γ_p, $\gamma_0 = 70$ and $k = 0.1$ are specified, resulting in a vague prior distribution, and $\alpha_0 = 1$. Additionally, a regular partition of the interval $(0, T)$ is used with $P = 8$ and $L = 8$.

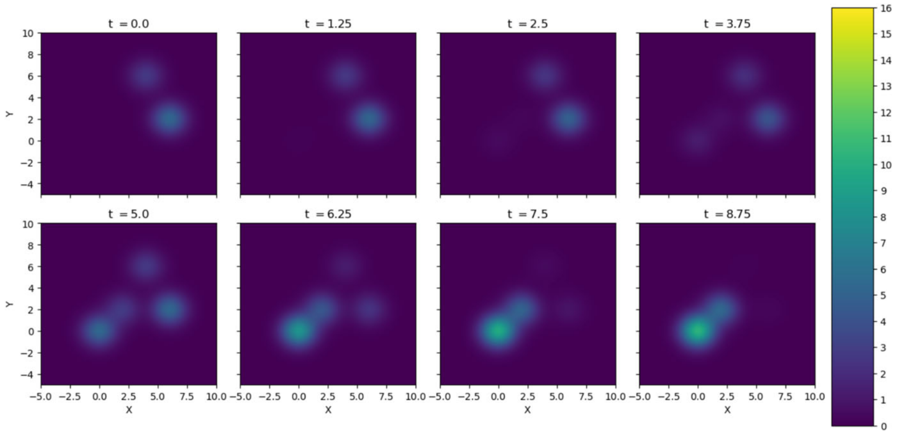

Fig. 2 λ evaluated at different times

Figure 2 shows the intensity function over the entire space at different times. This visual representation facilitates observation of spatial and temporal changes. The stochastic simulation of this NHPP was generated using the thinning algorithm (Lewis and Shedler 1979; Illian et al. 2008), which is summarized in Algorithm 1.

Algorithm 1 Simulation of a NHPP via thinning

Define

$$\lambda^* := \max_{(x,y)\in D,\ t\in(0,T)} \lambda(x, y, t)$$

Simulate a sample $\{(x_i^s, y_i^s, t_i^s)\}_{i=1}^{N^*}$ from a homogeneous Poisson process in $D \times (0, T)$ with intensity λ^* ▷ See Brémaud (2020) for details on HPP simulation

for $i = 1, \ldots, N^*$ **do**

  Compute $p_i = \dfrac{\lambda(x_i^s, y_i^s, t_i^s)}{\lambda^*}$

  Draw $u \sim \text{Uniform}(0, 1)$

  if $u \leq p_i$ **then**

    Accept (x_i^s, y_i^s, t_i^s)

  else

    Reject (x_i^s, y_i^s, t_i^s)

  end if

end for

return accepted samples

The simulated catalog, consisting of 727 events, and the code used to generate it are available at github.com/isaiasmanuel/NHPP.

Figure 3 presents the posterior mean of $\gamma_p f_p$, as defined in Eq. (5). In order to use information in the posterior distribution more effectively, Fig. 4 is also presented. Colors in Fig. 4 are the same as in Fig. 3, but transparency is modulated based on the

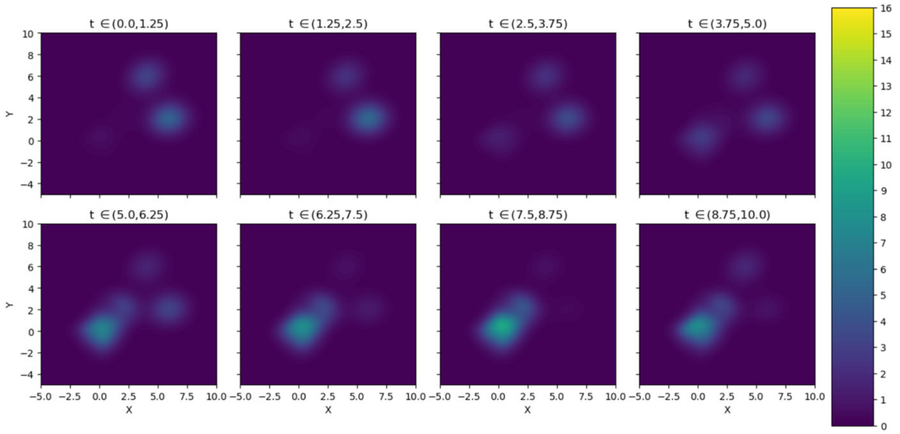

Fig. 3 Posterior mean of λ

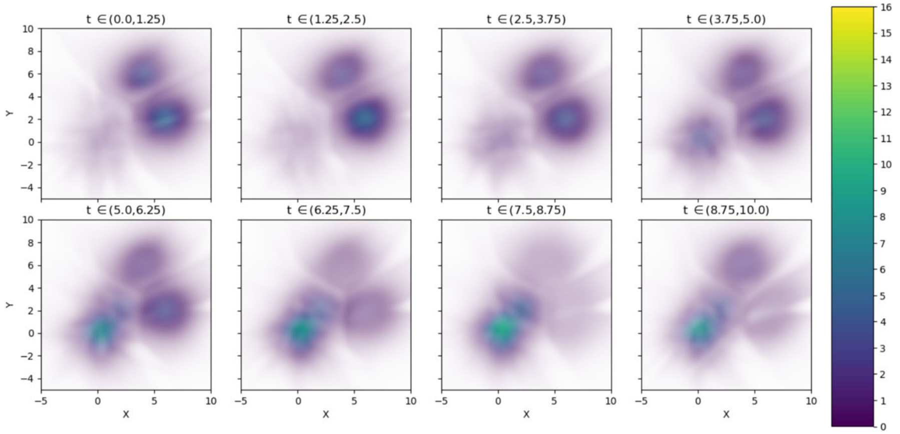

Fig. 4 Posterior mean of λ with transparency based on the coefficient of variation using simulated data

posterior coefficient of variation (CV), defined as

$$\mathrm{CV}_p(x, y) := \frac{\sigma(x, y)}{\mu(x, y)},$$

where μ and σ denote the posterior mean and variance of $\lambda(x, y, t)$ for $t \in (S_p, S_{p+1})$. The transparency for each figure p is then defined as

$$1 - \frac{\mathrm{CV}_p - \min_{x,y,p} \mathrm{CV}_p}{\mathrm{CV}_p},$$

In other words, if the posterior mean is high relative to the posterior standard deviation, the color is saturated. The transparency increases as the mean decreases relative to the standard deviation.

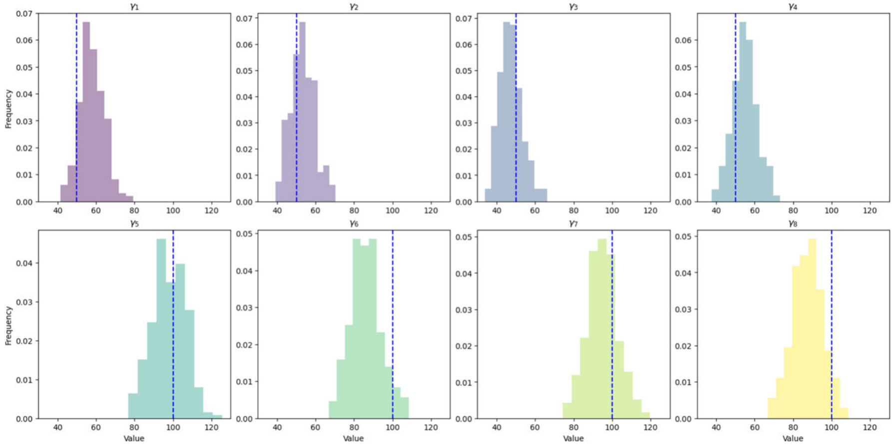

Fig. 5 Histogram of the posterior γ_p, $p = 1, 2, \ldots, P$ using simulated data; blue dashed lines are the real γ_p for each interval

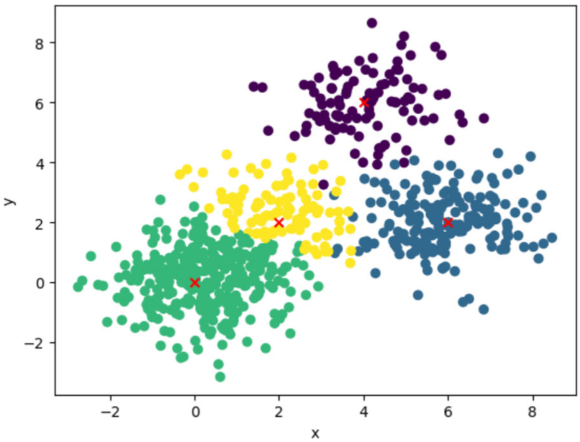

Fig. 6 Four-group clustering of the simulated data

As can be seen in Fig. 4, lower coefficient of variation values are observed in regions with higher values of λ (i.e., as expected, regions with the highest number of points are better estimated). In order to complement Fig. 4, Fig. 5 presents histograms of the marginal posterior samples for each γ_p along with the corresponding true values indicated by vertical dashed blue lines, according to Eq. (13). Figure 5 shows that the posterior distributions of γ_p adequately contain the true values for all $p = 1, 2, \ldots, P$.

In this work, inference on each element of the mixture is not the primary focus; instead, the objective is to infer the intensity function λ. Nevertheless, the label switching problem is observed, as expected in a finite mixture problem (Green 1995; Zhang et al. 2004; Bouguila and Elguebaly 2012). To address this, the algorithm proposed by Papastamoulis and Iliopoulos (2010) has been implemented. Relabeling is also performed according to lexicographic order of the cluster means to select the thinning in the chain. The discussion of burn-in and thinning of the chain is provided in the supplementary material. The code used in this section is available at github.com/isaiasmanuel/NHPP

To validate the proposed model, Fig. 6 presents the clustering of the observed points and the means of the Gaussian distributions (red crosses) used to define λ in Eq. (13). This clustering is obtained using the variation of information (VI) metric introduced by Meilă (2007) and the mcclust.ext library presented in Wade and Ghahramani (2018). The four mixture components are recovered adequately, even when the mixture weights change over time.

Clustering was also performed using Binder's loss, as proposed in Binder (1978); Dahl (2006). Nevertheless, a substantial overestimation of the number of clusters was observed, as reported in the supplementary material, which is consistent with the discussions in Wade and Ghahramani (2018); Dahl et al. (2022). For this reason, VI is used. Finally, readers primarily focused on clustering analysis can consult the papers Socher et al. (2011); Duan et al. (2024), where new priors for clusters are proposed based on distance measures and Bayesian spanning forests that rely on spectral clustering, which may be of interest. From this example, it is shown how GDP can be used to accurately estimate the intensity function of an inhomogeneous spatial Poisson process. Therefore, in the next section, the same methodology is applied to the Sawires et al. (2019) catalog to estimate the background intensity function.

3.2 Southern Mexico

As previously discussed in Sect. 1, several authors have pointed out that the background intensity function may change over time due to aseismic transients (Matthews and Reasenberg 1988; Marsan et al. 2013; Reverso et al. 2015; Nishikawa and Nishimura 2023) or strong earthquakes (Melgar et al. 2018; Cruz-Atienza et al. 2021). Nevertheless, a common assumption in the ETAS model is that the background intensity function is constant (Zhuang et al. 2002; Ross and Kolev 2022; Molkenthin et al. 2022). If this assumption is appropriate for the seismicity in southern Mexico, homogeneity is expected among all $\gamma_p f_p$ for $p = 1, 2, \ldots, P$. Alternatively, if the assumption is not appropriate, changes over space or time are expected in the regions where SSEs or strong earthquakes have been observed.

As reported by Radiguet et al. (2012, 2016) and Cruz-Atienza et al. (2021), SSEs in Guerrero, Mexico, between January 1, 2001, and December 31, 2015, exhibited an approximate periodicity of 4 years. For this reason, in the GDP inference for Mexican data, a regular time partition with $P = 4$ is used, resulting in an SSE in each time interval.

The epicenters taken from Sawires et al. (2019) are presented in Fig. 7 (colored according to their time interval p). The Middle America Trench (The General Bathymetric Chart of the Oceans 2015) is shown in fuchsia. Furthermore, the slip contour curves digitized from Radiguet et al. (2012) for the 2001–2002, 2006, and 2009–2010 SSEs are presented in blue, orange, and red, respectively (each 4 cm). Also, the 15 cm slip contour of the 2014 SSE is presented in black, which was digitized from Radiguet et al. (2016). Additionally, slip distributions, with 1-m resolution, are

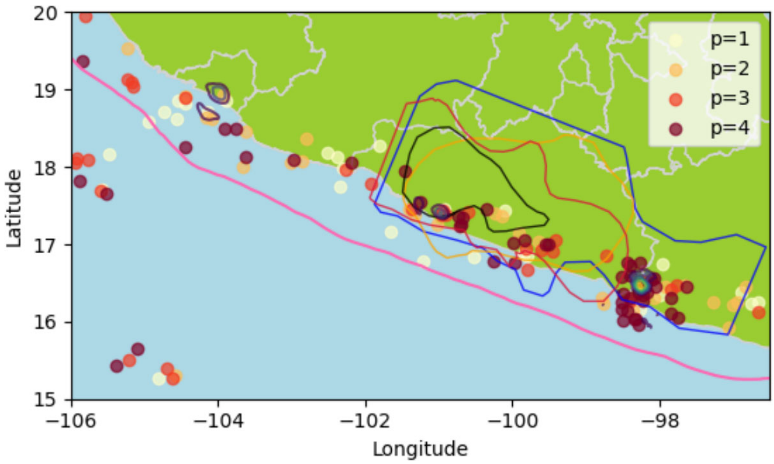

Fig. 7 Epicenters from the Sawires et al. (2019) catalog, SSE contours from Radiguet et al. (2012, 2016), slip distribution contours for earthquakes with $M \geq 7$ from U.S. Geological Survey (2017), and the Middle America Trench using data from The General Bathymetric Chart of the Oceans (2015)

presented for earthquakes that occurred on January 22, 2003, March 20, 2012, and April 18, 2014, with magnitudes 7.6, 7.4, and 7.2, and epicenters at -104.1040E° 18.770N°, -98.2310E° 16.493N°, and -100.9723E° 17.397N°, respectively, using data from U.S. Geological Survey (2017).

The hyperparameters of the NIW are taken as $\nu = 3$, $\eta = 0.01$, $\Sigma_0 = \mathbb{I}_2$, and $\mu_0 = (-102, 17)$, arbitrarily chosen to be a point near the center of the study region along the trench. The parameter μ is restricted to the observed domain, that is, $(-105.5\text{E}°, -96.5\text{N}°) \times (15\text{E}°, 19.5\text{N}°)$. Following the example in Sect. 3.1, $\alpha_0 = 1$ is taken. For γ_p, following Ross and Kolev (2022), $\gamma_0 = 1$ and $k = 0.01$ are used. Finally, $L = 12$ is set, similar to Chakrabarti et al. (2024), where $L = 10$ was used in a simulation example with a similar sample size. However, the MCMC was run with larger L values, but no improvements in the estimated intensities were observed. In contrast, the MCMC performance worsened due to label switching, as mixture components with zero observations were frequently allocated.

Figure 8 shows the posterior mean of $\gamma_p f_p$ for $p = 1, 2, 3, 4$. It is observed that the intensity function reaches its maximum near the slip distribution contours of strong earthquakes in the intervals $p = 1$ and $p = 4$. Furthermore, the background intensity function exhibits significant activity near the borders of the SSE slip regions. Similar behavior was previously reported by Fukuda (2018) for the Boso Peninsula, Japan. The increase in seismic activity at the borders of SSEs can be explained by stress redistribution caused by the largest SSE slips, which may reduce stress in some regions while increasing it in others, potentially triggering earthquakes in areas where stress has accumulated but not yet been released, as previously hypothesized for this region (Cruz-Atienza et al. 2021, 2025; Villafuerte et al. 2025).

While Fig. 8 shows the spatiotemporal changes, Fig. 9 presents the marginal posterior distributions of γ_p for $p = 1, 2, \ldots, 4$. In particular, a significant shift toward higher values is observed in the γ_4 distribution, indicating an increase in the expected

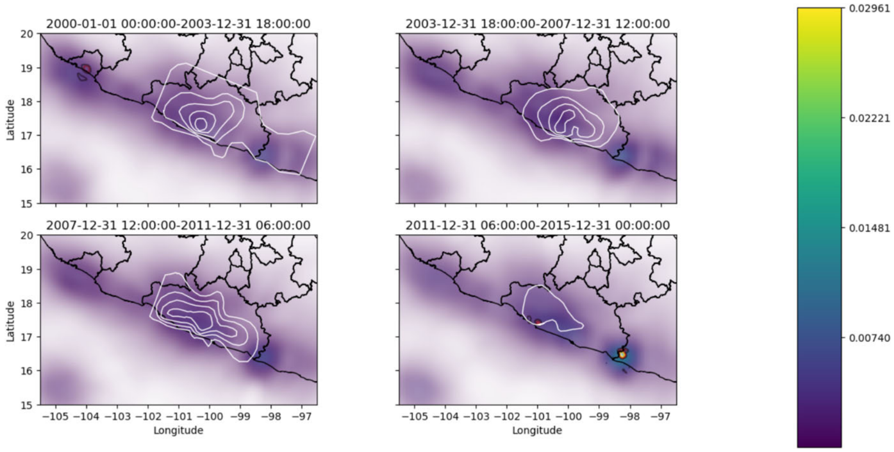

Fig. 8 Posterior mean of λ in $\frac{\text{events}}{\text{days deg}^2}$ with transparency based on the coefficient of variation using the catalog of Sawires et al. (2019); the gray curves in $p = 1, 2, 3$ are the slip contours at 4 cm intervals for the corresponding SSE in Fig. 7, and the curve in $p = 4$ is the slip contour at 15 cm

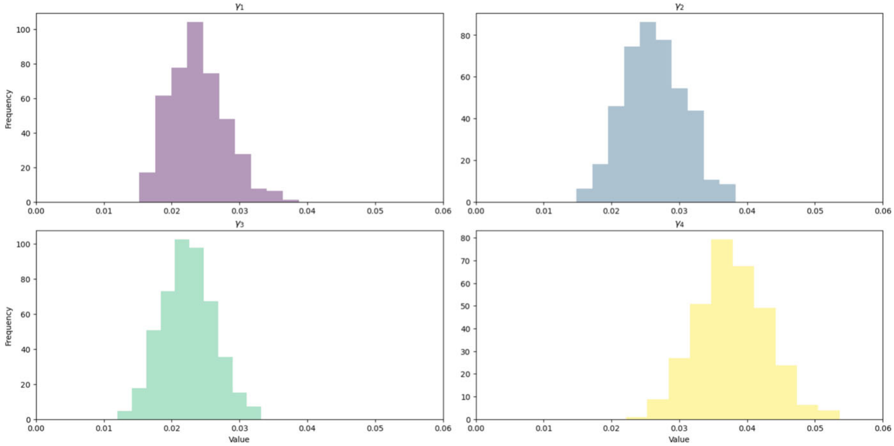

Fig. 9 Histogram of the posterior γ_p, $p = 1, 2, \ldots, P$ using the catalog of Sawires et al. (2019)

number of earthquakes during the last period, consistent with the occurrence of two strong earthquakes in this interval.

As presented in Sect. 3.1, one of the main advantages of using a DPM is that it allows clustering of events, as discussed in Müller et al. (2015), while estimating f_p. In Fig. 10, data from Sawires et al. (2019) is clustered into eight groups using the VI metric, as in Sect. 3.1. The clusters may be correlated with seismogenic features reported in the literature. For example, earthquakes assigned to the magenta cluster in Fig. 10 spatially correspond to the high b-value (1.50±0.10) region reported by Legrand et al. (2021). Specifically, earthquakes in the indigo cluster may be consistent with events associated with the ultra-slow (USL)-velocity layer reported by Song et al.

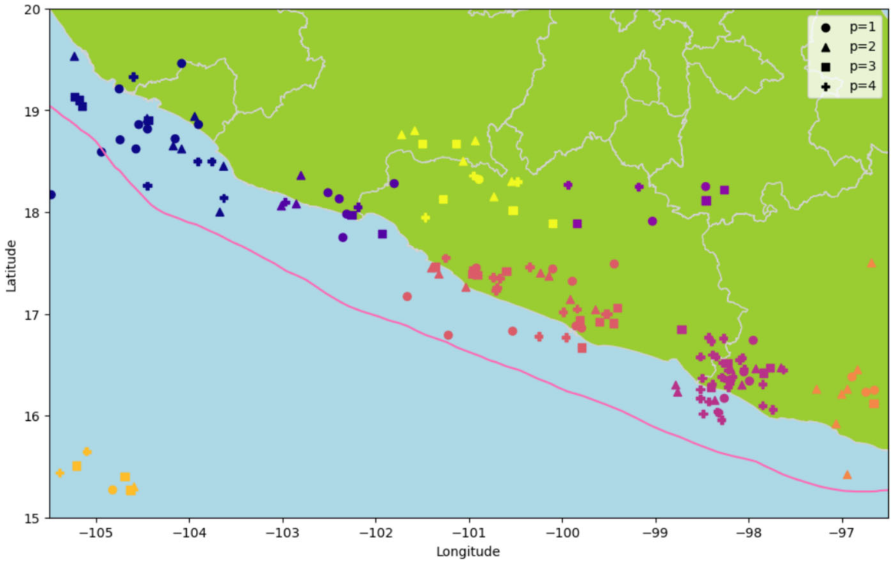

Fig. 10 Clustering into 8 groups using the catalog of Sawires et al. (2019); each color represents one group, and the geometric shapes represent the time period

(2009). Additionally, the blue cluster spatially corresponds to seismicity in the Rivera Plate subduction region.

Finally, the orange cluster corresponds to the earthquakes in the catalog that occurred in the oceanic crust of the Cocos Plate. The identification of this cluster was theoretically anticipated due to its geographical remoteness from other clusters, whereas for other clusters, their coincidence in shape with the subducted plate and the seismogenic zones is the salient feature to emphasize. The code to replicate this section is available at github.com/isaiasmanuel/NHPP, and details regarding burn-in and chain thinning can be found in the supplementary material.

4 Conclusions

Although the use of DPMs for nonparametric Bayesian estimation of the background intensity function was previously studied by Ross and Kolev (2022), only temporal homogeneity was considered. In this work, temporal changes in the background activity function are incorporated through GDP (Chakrabarti et al. (2024)). Furthermore, in the context of estimating the background intensity function with GDP, this approach allows the identification of seismogenic regions by relying solely on seismicity, a feature not previously discussed by Ross and Kolev (2022) and not intuitive to obtain using the approach presented in Molkenthin et al. (2022).

As discussed by Molkenthin et al. (2022), the Gaussian process requires specifying the correlation structure, whereas the DP approach requires determining the number of mixture components. In contrast, a DPM can capture a more flexible dependence structure than a Gaussian process, mainly due to its ability to model local behavior, unless heterogeneous correlations are considered, as recently discussed by (Muir and Ross 2023).

This work can be seen as an extension of the ideas presented in Matthews and Reasenberg (1988), which proposed the use of an inhomogeneous Poisson process for the background intensity function. The intensities estimated from the catalog by Sawires et al. (2019) for southern Mexico exhibit spatial and temporal changes associated with both strong regular earthquakes and SSEs. Therefore, the assumption of a homogeneous Poisson process is not realistic for southern Mexico.

Future work will investigate the self-exciting dynamics of earthquakes. While the present study focuses on inhomogeneities in the background intensity function, studies such as Ueda et al. (2021) and Briz-Redón and Mateu (2025) have shown that spatially or spatiotemporally varying self-excitation parameters can improve seismicity modeling. Additionally, it would be of interest to explore different DDP structures, such as higher-order autoregressive GDP, as well as considering the time-arrival structure introduced in Griffin and Steel (2006).

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Data Availability All codes required to replicate this work are publicly available.

Code availability All codes required to replicate this work are publicly available.

Declarations

Conflict of interest The authors declare that they have no conflict of interest.

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